

Anshil Gandhi

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EMPLOYMENT

Senior Software Engineer **Advanced Micro Devices Inc.** **May 2021 - contd.**

Full-time

- ❖ Successfully validated performance gains and ensured compatibility across diverse AI model architectures and hardware architectures.
- ❖ Addressed compiler regressions to enhance frameworks including triton, composable kernels, rocRAND, rccl, rocFFT, rocWaves, pytorch, tensorflow and onnxruntime.
- ❖ Participated in the advancements of LLVM by working on middleend and backend github issues.

Mentor **Outreachy** **May 2021 - Aug 2021**

Part-time

- ❖ Mentored an intern to implement GlobalSel instruction selector for the M68k backend in llvm.

Team Lead **Canadian Organization of Undergraduate Health Research** **June 2020 - March 2021**

Part-time

- ❖ Led a team to develop an Android application to collect data from patients.

Software Developer **NexOptic Technology** **April 2020 - Aug 2020**

Student intern

June 2019 - Aug 2019

- ❖ Implemented a Tensorflow based Image Signal Processor, which is an image pipeline of algorithms including bayer transformations and demosaicing, written in Python 3.
- ❖ Maintained and developed features for ALIIS™ on the Android platform in Kotlin.
- ❖ Developed an image streaming desktop application using dcraw, Spinnaker SDK, nuklear and OpenGL libraries, written in C++.
- ❖ Refactored ALIIS™ to process 720p video and improve frame rate by 50%, in C++.
- ❖ Implemented image file converters for PNG, TIFF and DNG in C++.

Student Intern **University of Alberta** **July 2016 - Aug 2016**

Full-time

- ❖ Implemented reinforcement learning environments for the game of 2048 and mountain car in Python 3.

EDUCATION

B.Sc. in Computing Science **University of Alberta** **Sept 2017 - June 2021**

and Mathematics

Relevant coursework Compiler Design, Machine Learning, Graph theory, GPU programming, Theory of computation, Computer organization and architecture, Algorithms and data structures, Operating Systems, Database Management, Reinforcement learning, Functional and logic programming, Multivariable calculus, Coding theory, Statistics.

PROJECTS

gazc is an LLVM-based compiler frontend for the Gazprea programming language, written in C++.

unixFS is a UNIX-based file system, written in C++.

Lianshell is a UNIX-based shell program, written in C++.

EulerTikz is an implementation of a force-based layout graph drawing algorithm, written in Python 3.

AWARDS

Open Kattis ranked 206 out of over 5000 problem solvers across the world

HackerRank 6-star gold badge in problem solving

Communitech's Code to Win challenge ranked among the top 75 coders across Canada

RMRC 2019 ranked third place in the ACM's regional programming contest out of 75 teams in the Rocky Mountain region

RMRC 2018 ranked 23rd place in the ACM's regional programming contest out of 65 teams in the Rocky Mountain region

TECHNICAL SKILLS

Programming Languages C/C++; HIP; CUDA; Python; Bash; Rust; Kotlin; Lisp

Tools and frameworks Android; UNIX; Git; Firebase; SQL; MongoDB